note

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1 Meta-learning

(Definition) Dataset

$$\mathcal{D} = \{(x, y)_i\}$$

(Definition) Task

$$\mathcal{T} = (\ell, \mathcal{D}^{\text{train}}, \mathcal{D}^{\text{test}})$$

Supervised learning (Probabilistic View)

$$\phi^* = \arg\max_{\phi} \log p(\phi|\mathcal{D})$$

Meta-supervised learning (Probabilistic View)

$$\phi^* = \arg\max_{\phi} \log p(\phi|\mathcal{D}, \mathcal{D}_{\text{prior}})$$

where $\mathcal{D}_{\text{prior}} = \{\mathcal{D}_1, \mathcal{D}_2, ..., \mathcal{D}_n\}$.

Meta-supervised learning (Probabilistic View, Approximated)

$$\phi^* = \arg\max_{\phi} \log p(\phi|\mathcal{D}, \theta^*)$$

$$\theta^* = \arg\max_{\theta} \log p(\theta, \mathcal{D}_{\text{prior}})$$

Meta-supervised learning (Mechanistic View)

$$\theta^* = \arg\min_{\theta} \frac{1}{M} \sum_{i=1}^{M} \ell_i(\phi_i, \mathcal{D}_i^{\text{test}})$$
$$\text{s.t.} \quad \phi^* = f_{\text{Adapt}}(\theta, \mathcal{D}_i^{\text{train}})$$

$$\theta^* = \arg\min_{\theta} \frac{1}{M} \sum_{i=1}^{M} \ell_i(\phi_i^*, \mathcal{D}_i^{\text{test}})$$
$$\text{s.t.} \quad \phi_i^* = f_{\text{Adapt}}(\theta, \mathcal{D}_i^{\text{train}})$$

$$\phi_i^* = f_\theta(\mathcal{D}_i^{\text{train}})$$
$$\phi_i^* = \arg\min_{\phi_i} \ell_i(\phi_i, \theta, \mathcal{D}_i^{\text{train}})$$

$$\theta^* = \arg\min_{\theta} \frac{1}{M} \sum_{i=1}^{M} \ell_i(\phi_i^*, \mathcal{D}_i^{\text{test}})$$
$$\text{s.t.} \quad \phi_i^* = \arg\min_{\phi_i} \ell_i(\phi_i, \theta, \mathcal{D}_i^{\text{train}})$$

$$\phi_i^{k+1} = \phi_i^k - \lambda \nabla_{\phi_i^k} \ell_i(\phi_i^k, \mathcal{D}_i^{\text{train}}), \quad \phi_i^0 = \theta$$

$$\begin{aligned} x_{1:T}^*, u_{1:T}^* = &\arg\min_{x_{1:T} \in \mathcal{X}, u_{1:T} \in \mathcal{U}} \sum_{t=1}^{T} C(x_t, u_t) \\ &\text{s.t.} \quad x_{t+1} = f(x_t, u_t), \quad x_1 = x_{\text{init}} \\ &\qquad\;\; x_{\min} \leq x_t \leq x_{\max}, \quad u_{\min} \leq u_t \leq u_{\max} \end{aligned}$$

$$\begin{aligned} \theta^* = \arg\min_{\theta} &\sum_{t=1}^{T} \|x_t^* - x_t^{\text{ref}}\|_2^2 + \|u_t^* - u_t^{\text{ref}}\|_2^2 \\ \text{s.t.} \quad x_{1:T}^*, u_{1:T}^* = &\arg\min_{x_{1:T} \in \mathcal{X}, u_{1:T} \in \mathcal{U}} \sum_{t=1}^{T} C_\theta(x_t, u_t) \\ &\text{s.t.} \quad x_{t+1} = f_\theta(x_t, u_t), \quad x_1 = x_{\text{init}} \\ &\qquad\;\; x_{\min} \leq x_t \leq x_{\max}, \quad u_{\min} \leq u_t \leq u_{\max} \end{aligned}$$

$$\theta^* = \arg\min_\theta \sum_{t=1}^{T} \|\hat{x}_t - x_t\|_2^2$$
$$\text{s.t.} \quad \hat{x}_t = f_\theta(x_{t-1}, u_{t-1})$$

$$\min_\theta \frac{1}{NMT}\left(\sum_{i=1}^{N}\sum_{j=1}^{M}\int_0^T \|x_{ij} - x_i^{\text{ref}}\|_2^2 + \mu_{\text{ctrl}}\|u_{ij}\|_2^2 dt + \mu_{\text{meta}}\|\theta\|_2^2\right)$$
$$\text{s.t.} \quad \dot{x}_{ij} = f(x_{ij}) + B(x_{ij})(u_{ij} + \hat{A}y(x_{ij};\phi))$$
$$u_{ij} = \bar{\pi}(x_{ij}, x_{ij}^{\text{ref}}, u_{ij}^{\text{ref}}; \kappa) - \hat{A}y(x_{ij};\phi)$$
$$\dot{\hat{A}} = \Gamma B(x)^T \nabla_{x_{ij}} \bar{V}(x_{ij}, x_{ij}^{\text{ref}}; \kappa) y(x;\phi)^T$$

$$\theta = (\phi, \Gamma, \kappa)$$
$$\min_\varphi$$

$$\dot{x} = f(x, u, \omega)$$
$$\dot{x} = f(x) + g(x)(u + w(x))$$

$$\|\omega\| \leq d$$
$$w(x) = \theta^T[\omega_1(x), \omega_2(x), ..., \omega_l(x)]$$
$$u = \pi(x)$$
$$u = \pi(x, \hat{\theta})$$
$$\dot{\hat{\theta}} = \rho(x)$$
$$\hat{\theta} \to \theta$$
$$\omega_i(x)$$

Definition

Dataset

$$\mathcal{D} = \{(x, y)_i\}$$

$$\theta^*$$
$$\mathcal{D}_i^{\text{train}}$$
$$\mathcal{D}_i^{\text{test}}$$
$$\mathcal{D}^{\text{train}}$$
$$\mathcal{D}^{\text{test}}$$